

DDL Command – ALTER & Schema Modification

1. WHAT IS THE ALTER COMMAND?

ALTER is a *Data Definition Language (DDL)* command used to modify the structure (schema) of an existing database object (tables, views, triggers, etc.) without re-creating it.

Command	Primary Purpose	Target State
CREATE	Creates a brand new structure/object	Object does not exist initially
ALTER	Modifies an existing structure/schema	Object must already exist in database

Core Operations Memory Trick

ALTER = Add + Drop + Modify + Rename + Constraints

2. MAJOR ALTER OPERATIONS & SYNTAX

A. ADD a Column / Multiple Columns ★

Adds new attribute fields to an existing relation. **Requirement:** You MUST specify the data type when adding a column.

Single Column Syntax

```
ALTER TABLE Employee
ADD Address VARCHAR2(30);
```

Multiple Columns Syntax

```
ALTER TABLE Employee
ADD (
    Address VARCHAR2(30),
    Age INT
);
```

B. DROP a Column ★★★

Removes an existing column from the schema.

```
ALTER TABLE Employee
DROP COLUMN Address;
```

Key Difference (ADD vs DROP Syntax)

- **ADD** requires Column Name + Data Type.
- **DROP COLUMN** requires Column Name ONLY (the DBMS already knows the stored data type).

C. MODIFY Column Data Type / Length ★

Changes the data type, precision, scale, or maximum character length of an existing attribute.

Change Data Type

```
ALTER TABLE Employee
MODIFY ID VARCHAR2(10);
```

Expand Character Length

```
ALTER TABLE Employee
MODIFY Name VARCHAR2(30);
```

D. RENAME Column vs. RENAME Table ★★

Renaming column identifiers vs. table identifiers requires distinct keyword placement:

Target	Syntax Command	Note / Keyword Requirement
Column	ALTER TABLE Employee RENAME COLUMN ID TO Roll_Number;	Uses RENAME COLUMN ... TO ...
Table	ALTER TABLE Employee RENAME TO Employee11;	Uses RENAME TO ...

3. ADDING CONSTRAINTS USING ALTER ★★★

Constraints (Primary Key, Foreign Key, NOT NULL, CHECK, UNIQUE) can be added to tables post-creation using ALTER.

```
ALTER TABLE Employee
ADD PRIMARY KEY (Roll_Number);
```

Prerequisite Condition

The target column (e.g., Roll_Number) **must already exist** in the table before attaching a constraint to it.

4. COMPLETE EXECUTION WALKTHROUGH & INSPECTION

Below is a step-by-step example showing the evolution of the schema through ALTER operations, checked with DESCRIBE:

Step-by-Step SQL Script

```
-- Step 1: Base Table Creation
CREATE TABLE Employee (
    ID NUMBER,
    Name VARCHAR2(20)
);

-- Step 2: Add Column
ALTER TABLE Employee
ADD Address VARCHAR2(30);

-- Step 3: Modify Data Type
ALTER TABLE Employee
MODIFY ID VARCHAR2(10);

-- Step 4: Rename Column
ALTER TABLE Employee
RENAME COLUMN ID TO Roll_Number;

-- Step 5: Add Constraint
ALTER TABLE Employee
ADD PRIMARY KEY (Roll_Number);

-- Step 6: Rename Table
ALTER TABLE Employee
RENAME TO Employee11;

-- Step 7: Inspect Schema
DESC Employee11;
```

Schema Output (DESC) Evolution

Stage	Active Table Structure
Initial	ID (NUMBER), Name (VARCHAR2(20))
After ADD	ID, Name, Address (VARCHAR2(30))
After MODIFY	ID (VARCHAR2(10)), Name, Address
After RENAME COL	Roll_Number, Name, Address
Final Table Name	Employee11 (old name Employee becomes invalid)

5. DDL COMMAND COMPARISON & COMMON EXAM TRAPS ⚠

Command	Action	Memory Mnemonic
CREATE	Defines a new structure/object	Make
ALTER	Modifies an existing structure	Change
DROP	Removes the structure/object entirely	Remove

Top 5 Exam Traps to Remember:

- ALTER is DDL:** It operates strictly on database *schema/structure*, not directly on data rows.
- ADD requires a datatype:** `ADD Address VARCHAR2(30);`
- DROP COLUMN omits datatypes:** `DROP COLUMN Address;`
- Different RENAME syntaxes:** Column uses `RENAME COLUMN`, Table uses `RENAME TO`.